

EXTREME ENVIRONMENTS

Survival against the odds

Polar areas and deserts are some of the least hospitable habitats on Earth. The lack of rain and extreme temperatures create difficult conditions for life, and the few humans that live there lead a seminomadic existence. Today, several of these fragile, untouched ecoregions are under threat because of the discovery of oil, gas, and other minerals.

Polar regions

Much of the Arctic and Antarctica is effectively a desert gripped by ice. Winters are long and permanently dark and summers are short but, because the Sun never sets, there is a continual source of energy for plant growth. Where rock is exposed, the soil is virtually nonexistent, and usually at freezing point or below. Trees cannot survive here, and vegetation is limited to mosses, lichens, fungi, and a handful of flowering plants. This open, rather featureless landscape, called tundra, is found between 60-80 degrees north and

south of the equator. It is far more extensive in the northern hemisphere, covering large tracts of northern Canada and Russia. Similar areas on mountains above the tree line are called alpine tundra.

Polar regions may support a number of large land animals, but most are reliant on the sea for their food. This is because despite the icy conditions, sea life is plentiful. The cold waters of the Arctic and Southern oceans are rich in oxygen and the seabed provides



**ALPINE
TUNDRA**

Found at altitudes of around 10,000 ft (3,000 m) with snow above and boreal forest below, alpine tundra is cold and windy and has sparse vegetation. Golden eagles hunt there during the summer as the open ground provides little cover for their prey.



**ARCTIC
TUNDRA**

The Arctic poppy is one of the few flowering plants found on the Arctic tundra. The short growing season means the plant must grow, flower, and produce its seeds rapidly. It is usually pollinated by flies, but can self-pollinate if necessary.



POLAR

Polar regions may seem inhospitable, but their icy waters support several specialist mammal species, including walruses and seals. Small ice floes and broken pack ice are ideal habitat for bearded seals, for example, as they need easy access to the water to feed.

ANTARCTICA ▷

Snow covers the tundra on the northeast tip of the Antarctic Peninsula in winter.

